

Chronological Sliding-Window Sequence Construction

Enforces Strict Isolation across Race, Driver, and Season Boundaries

Driver-Race Sequence Group: (Year, RoundNumber, Driver)

Sequence i (e.g., Lap 1 to 25):

20 Historical Encoder Laps: [t-19 ... t]
(Observed Telemetry, Weather, Laps, Rolling Stats)

5 Target Horizons: [t+1 ... t+5] (LapTime)

Sequence i+1 (Sliding window step +1 lap):

Historical Laps: [t-18 ... t+1]

Target Horizons: [t+2 ... t+6]

Strict Anti-Leakage Boundary Rule: Sequences never cross race or driver borders.
Tail windows with < 5 future laps in the same race are strictly discarded (zero zero-padding).